# Three colours and three generations from perturbativity in 3-3-1 models

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**Abstract**

In $SU(3)_C \times SU(3)_L \times U(1)_X$ ("3-3-1") gauge models, anomaly cancellation ties the number of fermion generations to the number of colours. We keep the number of colours $N_c$ general throughout, with every charge and anomaly coefficient computed as a function of $N_c$, and add a single requirement: no gauge coupling may become non-perturbative below the Planck scale. The $SU(3)_L$ anomaly together with asymptotic freedom of colour forces $N_c$ to be odd and $N_g = N_c$, which makes Witten's global anomaly condition automatic. $SU(3)_L$ running then excludes $N_c \geq 7$ and $U(1)_X$ running excludes $N_c = 5$, so that $N_c = N_g = 3$ is the only survivor, with quark charges $\frac{2}{3}$ and $-\frac{1}{3}$. The selection holds at two loops in the gauge couplings with two-loop Yukawa terms, with one-loop threshold matching across the heavy spectrum, for 3-3-1 breaking scales from 2 to 30 TeV, and in both variants $\beta = \pm 1/\sqrt{3}$. It takes the measured electroweak couplings at the TeV scale as input and does not predict them; in the $\beta = +1/\sqrt{3}$ variant the $N_c = 3$ margin is a factor of about 2 in the pole scale. Requiring the scalar sector itself to remain perturbative and bounded below up to $M_{\rm Pl}$ then bounds the exotic-quark masses relative to the bilepton mass. All numbers are reproduced by open code with regression tests.

## 1 Introduction

In 3-3-1 models [1, 2] the $SU(3)_L$ anomaly cancels only between generations, which links the number of generations to the number of colours: $N_g$ must be a multiple of $N_c$ [3, 4, 5]. With the number of light neutrinos taken from experiment, Frampton noted that this singles out $N_c = 3$ [6]. Here no generation or neutrino number is taken from experiment. We ask instead which $N_c$ is compatible with all gauge couplings remaining perturbative up to the Planck scale, given the measured electroweak couplings at the TeV scale. The answer is $N_c = N_g = 3$ only.

Section 2 sets up the model for general $N_c$. Section 3 gives the four-step selection; Sections 4–6 test its robustness, including one-loop threshold matching and the scalar sector. The conditions under which it holds are collected in Section 7, and Section 8 compares with prior work.

## 2 Setup for general $N_c$

The electric charge is $Q = T_3 + \beta T_8 + X$. The two variants without low-scale Landau poles are $\beta = \pm 1/\sqrt{3}$; the minimal $\beta = -\sqrt{3}$ model is excluded by atomic parity violation and sits at the LHC $Z'$ bound (Section 9). For colour group $SU(N_c)$ we take $N_g$ generations. Each has left-handed lepton triplets and quark triplets or anti-triplets, the third member of each quark multiplet being an exotic quark; $n$ of the $N_g$ quark generations sit in the anti-triplet.

Cancelling the perturbative anomalies, with the Yukawa couplings in place, gives

$$Q_u = \frac{N_c + 1}{2N_c}, \qquad Q_d = \frac{1 - N_c}{2N_c}, \tag{1}$$

which reduce to $\frac{2}{3}$ and $-\frac{1}{3}$ at $N_c = 3$. Every gauge anomaly ($X^3$, grav$^2X$, $SU(3)_L^2X$, $SU(N_c)^2X$, $SU(3)_L^3$) then vanishes identically in $N_c$, both in the Standard Model and in the full 3-3-1 content; this was checked in exact rational arithmetic for $N_c = 3, 5, 7$. Perturbative anomalies therefore fix the charges given $N_c$, but not $N_c$ itself.

The $U(1)_X$ charges for $\beta = -1/\sqrt{3}$ are $X = Q_u - \frac{1}{3}$ (quark triplet), $Q_d + \frac{1}{3}$ (anti-triplet), $-\frac{1}{3}$ (leptons), and $-\frac{1}{3}, +\frac{2}{3}, -\frac{1}{3}$ for the scalars $\eta, \rho, \chi$, fixed by the Yukawas and independent of $N_c$. For $\beta = +1/\sqrt{3}$: leptons $-\frac{2}{3}$, quark triplet $Q_u - \frac{2}{3}$, anti-triplet $Q_d + \frac{2}{3}$, scalars $\frac{1}{3}, \frac{1}{3}, -\frac{2}{3}$.

We require a non-abelian colour group, $N_c \geq 2$: for $N_c = 1$ there is no colour gauge field at all, and the asymptotic-freedom condition used below is vacuous.

## 3 The selection

**Step 1: $N_g = N_c$.** $SU(3)_L$ anomaly freedom requires equal numbers of triplets and anti-triplets,

$$N_g(1 + N_c) = 2nN_c. \tag{2}$$

For odd $N_c$, $\gcd(N_c, N_c + 1) = 1$, so $N_c \mid N_g$ and $N_g = kN_c$. With three quark flavours per generation, asymptotic freedom of $SU(N_c)$ requires $11N_c/3 - 2N_g > 0$, i.e. $k < 11/6$, so $k = 1$. For even $N_c$, $\gcd(N_c + 1, 2N_c) = 1$, so $2N_c \mid N_g$; but $2N_c > 11N_c/6$ violates asymptotic freedom. Hence $N_c$ is odd and $N_g = N_c$, with no further input. At $N_c = 3$ this reproduces the standard assignment of two quark generations in the anti-triplet and one in the triplet.

**Step 2: Witten's anomaly is implied.** In the Standard Model, Witten's global $SU(2)$ anomaly [7] requires an even number of doublets, $N_c + 1$ per generation. In 3-3-1, $SU(2)_L \subset SU(3)_L$ and $\pi_4(SU(3)) = 0$, so the UV group imposes no global anomaly of its own. Step 1 already forces $N_c$ odd, and the doublet count $N_g(1 + N_c) = 2nN_c$ is then even automatically. The spin–charge relation $(-1)^F = \exp(i\pi N_c(B - L))$ gives the same condition as a consequence rather than an input (all 29 solutions of (2) up to $N_c = 60$ checked).

**Step 3: $SU(3)_L$ running excludes $N_c \geq 7$.** With $N_g = N_c$ there are $N_c(1 + N_c)$ Weyl triplets. The one-loop coefficient $b_L = 11 - N_c(1 + N_c)/3 -$ (scalars), run from $1/\alpha_L(4\,\text{TeV}) = 31.5$, gives Table 1. Adding the minimal model's scalar sextet does not change the verdict.

| $N_c$ | $b_L$ (3 scalar triplets) | Landau pole |
|---|---|---|
| 3 | $+6.5$ | none (asymptotically free) |
| 5 | $+0.5$ | none |
| 7 | $-8.2$ | $1.3 \times 10^{14}\,\text{GeV}$, below $M_{\text{Pl}}$ |
| 9 | $-19.5$ | $10^8\,\text{GeV}$ |

Table 1: One-loop $SU(3)_L$ running with $N_g = N_c$.

**Step 4: $U(1)_X$ running excludes $N_c = 5$.** Two-loop gauge running from the 3-3-1 scale starts at $1/\alpha_X(4\,\text{TeV}) = 83.76$; we define "blow-up" as $g = 4\pi$. The results are in Table 2; the Planck scale is $1.2 \times 10^{19}\,\text{GeV}$. For $\beta = -1/\sqrt{3}$ a pole exactly at $M_{\text{Pl}}$ needs $a_X = 14.76$, so the one-loop margins are $+6.09$ for $N_c = 3$ and $-4.57$ for $N_c = 5$. Two-loop gauge terms move the blow-up scale by a factor of about 3.

| $N_c = N_g$ | one-loop $a_X$ ($\beta = -1/\sqrt{3}$; $+1/\sqrt{3}$) | blow-up, $\beta = -1/\sqrt{3}$ | blow-up, $\beta = +1/\sqrt{3}$ |
|---|---|---|---|
| 3 | $26/3 = 8.67$; 14.00 | $\mathbf{3.0 \times 10^{29}}$ GeV | $\mathbf{4.0 \times 10^{19}}$ GeV |
| 5 | 19.33; 27.33 | $1.3 \times 10^{15}$ GeV | $5.5 \times 10^{11}$ GeV |
| 7 | 34.44; 45.11 | $9.7 \times 10^{9}$ GeV | $3.1 \times 10^{8}$ GeV |

Table 2: Two-loop $U(1)_X$ blow-up scales.

**Two-loop Yukawa terms.** A Yukawa coupling $y\,\phi\psi_1\psi_2$ contributes $-g^3y^2\,n_{\text{terms}}\,[C_2(\psi_1) + C_2(\psi_2)]/d(G)/(16\pi^2)^2$ to each gauge $\beta$-function [8, 11]; this reproduces the Standard Model top-quark coefficients (17/10, 3/2, 2) exactly. We include the top Yukawa ($y_t = 0.83$ at 4 TeV) and one Yukawa $y_J$ for every exotic quark, and for every heavy lepton when $\beta = +1/\sqrt{3}$, held constant up to the blow-up. Real Yukawas decrease with energy, so this overstates their effect and makes the $N_c = 5$ exclusion conservative. Yukawas raise every pole by at most a factor of about 5 (Table 3), and no constant $y_J \leq 6$ lifts $N_c = 5$ to the Planck scale.

| $y_J$ | $\beta = -1/\sqrt{3}$: $N_c = 3$ / $N_c = 5$ | $\beta = +1/\sqrt{3}$: $N_c = 3$ / $N_c = 5$ |
|---|---|---|
| no Yukawas | $3.0 \times 10^{29}$ / $1.3 \times 10^{15}$ GeV | $4.0 \times 10^{19}$ / $5.5 \times 10^{11}$ GeV |
| 0 (top only) | $3.4 \times 10^{29}$ / $1.3 \times 10^{15}$ | $4.2 \times 10^{19}$ / $5.6 \times 10^{11}$ |
| 1 | $4.7 \times 10^{29}$ / $1.6 \times 10^{15}$ | $6.2 \times 10^{19}$ / $6.8 \times 10^{11}$ |
| 2 | $1.2 \times 10^{30}$ / $2.8 \times 10^{15}$ | $2.2 \times 10^{20}$ / $1.3 \times 10^{12}$ |

Table 3: Blow-up scales with two-loop Yukawa contributions.

## 4 Robustness

The two-loop machinery reproduces the Standard Model one-loop coefficients $(41/10, -19/6, -7)$ and all nine two-loop coefficients exactly. Table 4 varies the 3-3-1 scale and the perturbativity criterion.

| Check | $\beta = -1/\sqrt{3}$: $N_c = 3$ vs 5 | $\beta = +1/\sqrt{3}$: $N_c = 3$ vs 5 |
|---|---|---|
| 3-3-1 scale 2–30 TeV | $(2.7\text{–}3.7) \times 10^{29}$ vs $(0.8\text{–}4.5) \times 10^{15}$ | $(2.9\text{–}9.9) \times 10^{19}$ vs $(3.3\text{–}24) \times 10^{11}$ |
| criterion $\alpha = 1$ | $1.8 \times 10^{29}$ vs $1.0 \times 10^{15}$ | $2.9 \times 10^{19}$ vs $4.7 \times 10^{11}$ |
| criterion $g = 1$ (strict) | $5.3 \times 10^{25}$ vs $2.7 \times 10^{13}$ | $2.0 \times 10^{17}$ vs $3.6 \times 10^{10}$ ($N_c = 3$ fails) |

Table 4: Blow-up scales in GeV under variations.

For $\beta = -1/\sqrt{3}$, $N_c = N_g = 3$ is selected under every criterion, with its blow-up 6–10 orders of magnitude above $M_{\text{Pl}}$ while $N_c = 5$ fails 4–6 orders below. For $\beta = +1/\sqrt{3}$, 3 is selected under the standard criteria, but its pole lies only a factor 2–3 above $M_{\text{Pl}}$ without Yukawas (3–5 with them); under the strict $g = 1$ criterion, $N_c = 3$ fails as well.

## 5 One-loop threshold matching

The breaking $SU(3)_L \times U(1)_X \to SU(2)_L \times U(1)_Y$ is matched at one loop in $\overline{\text{MS}}$ [12, 13]:

$$\frac{1}{\alpha_{\text{low}}(\mu)} = \frac{1}{\alpha_{\text{high}}(\mu)} - \frac{\lambda}{12\pi}, \quad \lambda = \sum_V t^2\Big(1 - 21\ln\frac{M_V}{\mu}\Big) + 8\sum_{\text{Dirac}} t^2 \ln\frac{M_F}{\mu} + \sum_{\text{real scalars}} t^2 \ln\frac{M_S}{\mu}, \quad (3)$$

with $1/\alpha_X + \beta^2/\alpha_L$ as the high-energy side for $U(1)_Y$. The heavy spectrum is: one complex $SU(2)$ doublet of bilepton vectors ($|Y| = \frac{1}{2}$, mass $M_V$); the $Z'$ ($M_{Z'} = 1.225\,M_V$); $N_c$ exotic

quarks (plus $N_c$ heavy leptons for $\beta = +1/\sqrt{3}$) of mass $r_J M_V$, whose Yukawa $y_J = r_J g_L/\sqrt{2}$ enters at two loops; and nine heavy real scalars of mass $r_S M_V$ (a second doublet, a charged singlet and three neutral fields), with the option of a light second doublet.

As a check, at one loop the $N_c = 3$ pole must not depend on the matching scale once the spectrum is fixed: moving $\mu$ from $M_V/3$ to $3M_V$ changes ln(pole) by less than $10^{-9}$, against a factor 2.6 without thresholds. $N_c = 5$ cannot pass this check, because the low-energy theory is the three-colour Standard Model; its pole moves by a factor 1.5 with $\mu$, and we take its most favourable $\mu$.

Scanning $r_J = 0.5$–$3$, $r_S = \frac{1}{3}$–$3$, $M_V = 2$–$10\,\text{TeV}$ and a heavy or light second doublet (two-loop running, Yukawas included) gives Table 5. The worst case for $N_c = 3$ (lightest exotics, light scalars, light second doublet, $M_V = 2\,\text{TeV}$) puts the $Z'$ at 2.45 TeV, far below the LHC limits of Section 9. Imposing the most conservative limit from our recast of the full Run-2 dilepton search (Appendix A), $M_{Z'} \geq 4.6\,\text{TeV}$ for $\beta = +1/\sqrt{3}$ and 4.0 TeV for $\beta = -1/\sqrt{3}$, restricts the scan to $M_V \geq 3.76$ and 3.27 TeV respectively. At two loops, moving $\mu$ from $M_V/2$ to $2M_V$ changes the pole by 1%.

| | $N_c = 3$, worst case | $N_c = 5$, best case | verdict |
|---|---|---|---|
| $\beta = -1/\sqrt{3}$ | $2.4 \times 10^{29}\,\text{GeV}$ $(2 \times 10^{10} M_{\text{Pl}})$ | $7.2 \times 10^{15}\,\text{GeV}$ | only $N_c = 3$ |
| $\beta = +1/\sqrt{3}$ | $2.0 \times 10^{19}\,\text{GeV}$ $(1.64\, M_{\text{Pl}})$ | $3.8 \times 10^{12}\,\text{GeV}$ | only $N_c = 3$ |
| $\beta = +1/\sqrt{3}$, $M_{Z'} \geq 4.6\,\text{TeV}$ (LHC) | $2.17\, M_{\text{Pl}}$ | — | only $N_c = 3$ |

Table 5: Selection with one-loop threshold matching over the scanned spectrum. The last row restricts the scan to the dilepton bound of Appendix A.

The selection holds for any cutoff $\Lambda$ between the best $N_c = 5$ pole and the worst $N_c = 3$ pole. Over all criteria ($g = 4\pi$ or $\alpha = 1$, with or without Yukawas), any $\Lambda$ from $7.2 \times 10^{15}$ to $1.4 \times 10^{19}\,\text{GeV}$ works. This window contains $M_{\text{Pl}}$, the reduced Planck mass and the conventional unification scale.

## 6 The scalar sector to the Planck scale

The selection requires every coupling of 3-3-1 at $N_c = 3$ to stay perturbative to $M_{\text{Pl}}$, so the scalar sector must pass the same test. We implement the general one-loop quartic and Yukawa RGEs [9, 10, 11] on the 18 real scalars; the engine reproduces the Standard Model $\beta_\lambda$ and $\beta_{y_t}$ exactly. The nine quartics allowed by the model's discrete symmetry close under the RGEs (residual $10^{-15}$), and the potential is gauge invariant to $10^{-16}$. The light-Higgs quartic is matched at 4 TeV to $\lambda_{\text{SM}} = 0.076$ at tree level in the decoupling limit. Survival means every $|\lambda| < 4\pi$, every Yukawa $< \sqrt{4\pi}$, and a potential bounded below, checked numerically at eight scales up to $M_{\text{Pl}}$.

3-3-1 at $N_c = 3$ survives in both variants, in a corner of parameter space: 17 of 250 viable points ($\beta = -1/\sqrt{3}$) and 13 of 250 ($\beta = +1/\sqrt{3}$) under small priors, none with $O(1)$ quartics. In the survivors the extra scalars cure the Standard Model's metastability. Consistency to $M_{\text{Pl}}$ then requires the exotic Yukawa at 4 TeV to be below 0.80 ($\beta = -1/\sqrt{3}$) or 0.62 ($\beta = +1/\sqrt{3}$), i.e. exotic-quark masses below $1.79\, M_V$ or $1.40\, M_V$; the top's doublet must carry $c^2 \gtrsim 0.68$ of the electroweak vev squared; and all quartics at 4 TeV must be $\lesssim 0.37$. These statements are one-loop, with tree-level matching, and the search is a random sampling rather than a proof that the surviving region is unique.

## 7 Conditions and caveats

1. *The framework.* Everything assumes 3-3-1 with $\beta = \pm 1/\sqrt{3}$.

2. *Perturbativity to the Planck scale* is a standard criterion, not a theorem. A pole below $M_{\rm Pl}$ signals new physics rather than an inconsistency.

3. *Measured couplings are input.* Every $N_c$ is run from $\sin^2\theta_W = 0.2504$, $1/\alpha_{\rm em} = 125.7$ and $\alpha_s$ at 4 TeV; in 3-3-1, $1/\alpha_X = (1 - 4s^2/3)/\alpha_{\rm em}$. For $\beta = -1/\sqrt{3}$, $N_c = 3$ reaches $M_{\rm Pl}$ for $\sin^2\theta_W < 0.449$, while $N_c = 5$ does so only for $0.066 < \sin^2\theta_W < 0.074$, a factor 3.1–3.4 below the measured value. For $\beta = +1/\sqrt{3}$, $N_c = 5$ fails at every weak angle, and $N_c = 3$ survives for $\sin^2\theta_W < 0.266$ (tree-level matching; +0.010 to +0.035 in margin with thresholds). The precise statement is therefore: *given the measured electroweak couplings*, only $N_c = 3$ stays perturbative to $M_{\rm Pl}$.

4. *Minimality is not explained.* Steps 1–2 leave $\{3, 5, 7, \dots\}$; the running does the selecting.

5. *No numbers follow.* The structure adds about eleven scalar parameters beyond the Standard Model's two and blocks gauge unification ($SU(3)_C$ and $SU(3)_L$ never meet).

## 8 Relation to prior work

That $N_g$ is a multiple of $N_c$ in 3-3-1 is known [1, 2, 3, 4]. Frampton obtained $N_c = 3$ by inputting three light neutrinos [6]. Charge quantization for arbitrary $N_c$ was discussed in [14], generation counting for arbitrary $\beta$ in [15], and the $N_c$-dependence of the charges in [16]. The closest predecessor is Doff and Pisano [5], who keep $n_c$ general and find $N_g = k\, n_c$ for any $k \geq 1$; they do not use asymptotic freedom to set $k = 1$, nor running to choose $n_c$. Landau poles in 3-3-1 at $N_c = 3$ are studied in [17, 18, 19].

The step taken here is to keep $N_c$ general throughout (charges, anomalies, scalars, two-loop running, thresholds, scalar sector) without taking the number of generations or neutrinos from experiment. Searches of arXiv and INSPIRE, including every paper citing the founding 3-3-1 papers, Frampton [6] and Doff–Pisano [5], found no prior use of asymptotic freedom plus running to fix $N_c$. This is not proof of novelty, and comments from experts are welcome.

## 9 Experimental tests

The $\beta = \pm 1/\sqrt{3}$ models predict a $Z'$, singly or doubly charged bilepton gauge bosons and exotic quarks, which the LHC and future colliders will find or exclude. Dilepton limits on the $Z'$ depend on the variant and on its decays. Published recasts use partial data: for $\beta = -1/\sqrt{3}$, $M_{Z'} > 4$ TeV from 36.1 fb$^{-1}$ at 13 TeV with Standard Model decays only [20], weakened to 2.64 TeV with decays into exotic fermions open [21]; for $\beta = +1/\sqrt{3}$, the most recent dedicated recast we found uses 8 TeV data and gives $M_{Z'} \gtrsim 2.9$ TeV [22]. Appendix A recasts the ATLAS search with the full 139 fb$^{-1}$ [26]:

$$\beta = +1/\sqrt{3}:\ M_{Z'} > 5.0\text{–}5.2\ \text{TeV}\ (4.6\ \text{TeV}), \qquad \beta = -1/\sqrt{3}:\ M_{Z'} > 4.7\text{–}5.0\ \text{TeV}\ (4.0\ \text{TeV}),$$

for decays to Standard Model fermions only (in brackets: the lowest value with every exotic fermion and every scalar-triplet component open and massless). Since $M_V = M_{Z'}/1.225$, the bilepton gauge bosons then lie above about 3.3–4.2 TeV, and the bounds of Section 6 allow exotic quarks up to 1.40–1.79 $M_V$, so they may or may not be light enough for the $Z'$ to decay into.

Consistency to $M_{\rm Pl}$ adds the relations of Section 6: exotic quarks lighter than 1.79 $M_V$ or 1.40 $M_V$, $M_{Z'} = 1.225\, M_V$, and the top's doublet carrying $\gtrsim 70\%$ of the electroweak vev squared. A discovery violating these would break the selection without excluding 3-3-1.

In the minimal $\beta = -\sqrt{3}$ model, $\sin^2\theta_W$ reaches $\frac{1}{4}$ at 3.68 TeV at one loop, where $g_X$ has its Landau pole, so its $Z'$ must lie below about 3.7 TeV. The LHC bound in this model is $M_{Z'} > 3.7$ TeV (13 TeV, 36.1 fb$^{-1}$, exotic decays included) [23], and atomic parity violation

in caesium requires $M_{Z'} \gtrsim 4.2$–$5.4\,\text{TeV}$ [24]. The minimal model is therefore excluded unless higher-order effects raise its pole by more than about 14%.

Finally, $N_c$ can be checked directly at low energy. With the $N_c$-dependent charges of Section 2 and the $O(\alpha_s)$ correction, $R = \sigma(e^+e^- \to \text{hadrons})/\sigma(e^+e^- \to \mu^+\mu^-)$ below charm is $N_c(Q_u^2 + 2Q_d^2)(1 + 3C_F\alpha_s/4\pi)$, which is 2.173 at $N_c = 3$. KEDR's weighted average over 1.84–3.05 GeV is $R = 2.225 \pm 0.020 \pm 0.047$ [25], so $N_c = 3$ agrees to $1.0\sigma$, while $N_c = 2$ and 5 miss by $15\sigma$ and $33\sigma$. This check is independent of the running argument; it is of course not new, but it confirms that the $N_c$-dependent charges reproduce the measured value.

# A Recast of the ATLAS 139 fb$^{-1}$ dilepton search

ATLAS published 95% CL upper limits on the fiducial $\sigma \times B(X \to \ell\ell)$, for one lepton flavour, for relative widths $\Gamma/M = 0$–10% [26]. We use the official HEPData tables [27] for the observed and expected limits of the combined $ee + \mu\mu$ channel. The fiducial $Z'_{\text{SSM}}$ prediction is reconstructed from the tabulated ratio of the observed limit to the SSM cross-section; it crosses the 3%-width limit at 5.08 TeV (published: 5.1 TeV). The tables label the limits in pb, but their values are the femtobarn numbers quoted in the paper (3.6 fb at 250 GeV, 0.014 fb at 6 TeV), and we use them in fb. ATLAS's $Z'_\psi$ and $Z'_\chi$ predictions are not tabulated; for the validation below we digitised them from the vector figures of the paper. The same digitisation of the limit curves reproduces the HEPData tables to a uniform +2.7% with at most 1.2% scatter, and we correct the two model curves for that offset.

**Method.** In the narrow-width approximation, $\sigma B \propto \sum_q L_{q\bar{q}}(M)\,(g_{L,q}^2 + g_{R,q}^2) \times B(\ell\ell)$. We take the ATLAS $Z'_{\text{SSM}}$ prediction and multiply it by the ratio of this quantity between the 3-3-1 $Z'$ and $Z'_{\text{SSM}}$. Higher-order QCD corrections and the fiducial acceptance, the same for vector resonances to a few per cent, cancel in the ratio. Parton luminosities at $\mu = M$ use the PDF4LHC21 and NNPDF3.1 NNLO central sets. The mass limit is where the prediction crosses the observed limit at the model's width, interpolated between the published widths.

**Couplings.** With $X = Y - \beta T_8$, the heavy neutral boson orthogonal to hypercharge couples to each chiral field as

$$g_{Z'}(f) = \frac{g\,(T_8 - \beta\, t_W^2\, Y)}{\sqrt{1 - \beta^2 t_W^2}}, \tag{4}$$

with $T_8 = \pm 1/(2\sqrt{3})$ for SM doublets in triplets and anti-triplets and $T_8 = 0$ for singlets. This agrees with a direct diagonalisation to $10^{-16}$, reproduces the known $\beta = -1/\sqrt{3}$ electron vector coupling $g(1-4s_W^2)/(4c_W\sqrt{3-4s_W^2})$, and for $\beta = -\sqrt{3}$ diverges at $s_W^2 = \frac{1}{4}$, the minimal model's Landau pole. Leptons sit in triplets; two quark generations sit in anti-triplets and one in the triplet. We take either the third (assignment A, the usual choice) or the first generation (B) in the triplet. Couplings are evaluated with $s_W^2 = 0.2504$ (4 TeV) and, as a check, 0.2312.

**Validation.** The procedure reproduces the benchmark widths (2.99, 0.53 and 1.16% for SSM, $\psi$, $\chi$, against ATLAS's 3, 0.5 and 1.2%). Applied to the $Z'_{\text{SSM}}$ prediction with the $E_6$ couplings, it reproduces ATLAS's own $Z'_\psi$ and $Z'_\chi$ predictions to within 3 and 4% between 3 and 5 TeV with PDF4LHC21, and their mass limits: 4.53 and 4.81 TeV against the published 4.5 and 4.8 TeV. With NNPDF3.1 the $\chi$ limit comes out at 4.96 TeV; the difference traces to the large-$x$ antiquarks, where the two sets differ, so we use PDF4LHC21 as the baseline and quote the NNPDF3.1 spread. Opening the exotic decays for $\beta = -1/\sqrt{3}$ reduces $B(\ell\ell)$ to 37% of its SM-only value, consistent with the 34% found independently in Ref. [21].

**Independent reproduction.** We repeated the calculation with PYTHIA 8's own $Z'$ process [28] (`ffbar2gmZZprime`, $Z'$ alone), which shares only the couplings with the method above: its own LO PDF (NNPDF2.3), width calculation and full Breit–Wigner line shape. The ratio of $\sigma B(ee)$ to the SSM comes out 4–8% lower, consistent with the LO PDF and with off-shell tails that favour the wider SSM. Taken through the same ATLAS limits it gives $M_{Z'} > 5.16$ and 5.00 TeV for $\beta = +1/\sqrt{3}$ (assignments A, B) and 4.75 TeV for $\beta = -1/\sqrt{3}$ (A), within 30–40 GeV of Table 6.

**Decays, interference and mixing.** The bilepton-pair channel is closed, since $M_{Z'} = 1.225\,M_V < 2M_V$, and the $Z'$ has no tree-level $W$–bilepton coupling (the $SU(3)$ structure constants $f_{8ab}$ with $a \in \{1, 2\}$ vanish). Scalar decays are bounded by treating every component of the three scalar triplets as a massless complex scalar, which overcounts heavy states and eaten Goldstones and, by the equivalence theorem, contains the $W_L W_L$ and $Zh$ channels opened by $Z$–$Z'$ mixing. This "all open" case lowers the limits by at most 0.2 TeV beyond the exotic-fermion case. Interference with $\gamma/Z$ Drell–Yan, from the LO line shape, changes the signal within ±10% of the peak by −3.3 to +1.6% (±0.7% within ±2%); the SSM benchmark has interference of the same sign and size (−4.7%), so it largely cancels in the ratio. A 3% change in $\sigma B$ moves the limit by about 15 GeV. A $Z$–$Z'$ mixing angle of $10^{-3}$, three times its natural size $M_Z^2/M_{Z'}^2$, changes $\sigma B$ by at most 0.4%. Not included are differences in higher-order corrections between models.

| $\beta$ | assignment | decays | $\Gamma/M$ | $B(ee)$ | observed | expected |
|---|---|---|---|---|---|---|
| $+1/\sqrt{3}$ | A | SM | 1.68% | 6.58% | 5.21 TeV | 5.18 TeV |
| | A | SM + exotic | 3.27% | 3.38% | 4.89 TeV | 4.86 TeV |
| | A | all open | 3.64% | 3.04% | 4.83 TeV | 4.80 TeV |
| | B | SM | 1.68% | 6.58% | 5.03 TeV | 5.00 TeV |
| | B | SM + exotic | 3.27% | 3.38% | 4.71 TeV | 4.67 TeV |
| | B | all open | 3.64% | 3.04% | 4.64 TeV | 4.61 TeV |
| $-1/\sqrt{3}$ | A | SM | 1.15% | 3.85% | 4.79 TeV | 4.75 TeV |
| | A | SM + exotic | 3.09% | 1.43% | 4.05 TeV | 4.24 TeV |
| | A | all open | 3.46% | 1.28% | 3.99 TeV | 4.17 TeV |
| | B | SM | 1.15% | 3.85% | 4.96 TeV | 4.93 TeV |
| | B | SM + exotic | 3.09% | 1.43% | 4.39 TeV | 4.43 TeV |
| | B | all open | 3.46% | 1.28% | 4.19 TeV | 4.37 TeV |

Table 6: 95% CL lower limits on $M_{Z'}$ from the ATLAS 139 fb$^{-1}$ dilepton search (HEPData limits, PDF4LHC21, $s_W^2 = 0.2504$). A: third quark generation in the triplet; B: first. "SM + exotic": decays to the exotic quarks and heavy leptons open and massless; "all open": every scalar-triplet component as well. Over the $s_W^2$ scheme and the PDF set the limits move by up to 0.05 TeV.

## Reproducibility

Each number above is produced by a script in `src/m331_*.py`, with reference output in `results/m331/` and regression tests in `tests/test_331_*.py`, run on every commit. The recast is `src/m331_zprime_recast.py`; the HEPData tables, the digitised $\psi/\chi$ curves, the parton luminosities and the PYTHIA cross-check it reads are produced by the scripts in `tools/`.

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